

Arctic Ice Sheet Dynamics: Observations from 2019-2024

Executive Summary

This report presents five years of satellite and field observation data documenting changes in Arctic ice sheet volume, extent, and seasonal melt patterns. Data was collected in partnership with the Norwegian Polar Institute and MIT Climate Lab.

1. Background

The Arctic is warming at approximately four times the global average rate, a phenomenon known as Arctic amplification. This accelerated warming has significant implications for global sea level rise, ocean circulation patterns, and weather systems across the Northern Hemisphere.

2. Observations

Between 2019 and 2024, average summer sea ice extent declined by 12.4% compared to the 1981-2010 baseline. The Greenland Ice Sheet lost an estimated 280 billion tonnes of ice annually, contributing approximately 0.7mm per year to global sea level rise. Permafrost thaw was recorded at depths previously considered stable.

3. Modelling Projections

Under the SSP2-4.5 emissions scenario, summer Arctic sea ice is projected to reach near-zero extent by 2060. Under SSP5-8.5, this threshold may be crossed as early as 2035. Both scenarios predict increased frequency of polar vortex disruptions affecting mid-latitude weather patterns in Europe and North America.

4. Policy Recommendations

Immediate reduction of black carbon emissions from shipping and industry in the Arctic region is recommended as a short-term mitigation measure. Long-term strategies must include binding international agreements on methane emissions from permafrost regions and accelerated transition away from fossil fuel dependency.